MZ505 Inventory Systems — Formula Sheet

0 · Which model?

	EOQ / EPQ	Newsvendor	Order-up-to
Review	continuous	single period	periodic
Demand	deterministic, constant	random	random, stationary
Unmet demand	none allowed	lost sales	backordered
Setup cost	yes (S)	no (or fixed K)	0 (or $K \to$ sets T)
Obsolescence	never	after the season	never
Replenishments	unlimited	one	unlimited
Trade-off	setup vs. holding	too much vs. too little	holding vs. backorder
Lead-time demand	ROP $= DL$	–	covers $L+1$ periods
Decision	Q and ROP	Q once	S (and T)
Buffer	none	$z\sigma$ of season	$\approx z\sigma_{L+1}$

Clues: constant annual demand, cost per order, "how much / when" → EOQ · production rate P, setup → EPQ · price table by order size → all-unit discounts · several products on one truck, common + specific order cost → aggregation · intervals 1, 2, 4, 8 weeks → power of two · one season, perishable/fashion, order before demand, salvage, lost sales → newsvendor · supplier + retailer, w, returns → contracts · periodic review, lead time, demand per period, backorders, in-stock/fill rate, inventory position → order-up-to.
Newsvendor vs. OUT: newsvendor leftovers lose value (s) and shortages are lost; OUT leftovers carry over (cost h) and shortages wait. So OUT protects $L+1$ periods, newsvendor only the season.

Units: $D_{\text{yr}} = 52D_{\text{wk}} = 12D_{\text{mo}} \cdot H = iC$ (€/unit/yr) · P, D, L in the same unit · aggregate k periods: $\mu_k = k\mu$, $\sigma_k = \sqrt{k}\sigma$ · split into k sub-periods: μ/k, $\sigma/\sqrt{k}$ (month → week: $\mu/4.33$, $\sigma/\sqrt{4.33}$) · annual $H \to$ weekly $H/52$.

1 · EOQ and EPQ

Variables: D demand rate · S cost per order/setup · C unit cost · i holding rate (%/yr) · $H = iC$ holding cost per unit per year · Q lot size · P production rate ($P > D$) · L lead time · T cycle · n orders (runs) per year · T_p uptime · T_d downtime · TVC total variable cost/yr · TC total cost/yr.

EOQ assumptions: demand & lead time known and constant; instantaneous receipt; constant price; holding on average inventory; constant order cost; no backorders. Trade-off: ordering (economies of scale) vs. holding. **EPQ:** production at P while demand consumes at $D \to$ stock rises at $P - D$, never reaches Q. EPQ > EOQ; $P \to \infty$ gives EOQ. Changes vs. EOQ in red.

	EOQ	EPQ
TVC(Q)	$S\frac{D}{Q} + H\frac{Q}{2}$	$S\frac{D}{Q} + H\frac{Q}{2}\left(1 - \frac{D}{P}\right)$
TC(Q)	$CD + \text{TVC}(Q)$	
FOC	$-S\frac{D}{Q^2} + \frac{H}{2} = 0$	$-S\frac{D}{Q^2} + \frac{H}{2}\left(1 - \frac{D}{P}\right) = 0$
Q^*	$\sqrt{\frac{2DS}{H}}$	$\sqrt{\frac{2DS}{H(1-D/P)}}$
TVC(Q^*)	$\sqrt{2DSH}$	$\sqrt{2DSH(1-D/P)}$
$I_{\max}$	Q	$T_p(P-D) = Q(1-D/P)$
$\bar{I}$ avg	$Q/2$	$I_{\max}/2$ (not $Q/2$)
T_p	0	Q/P
T_d	Q/D	$I_{\max}/D$
T	Q/D	$T_p + T_d = Q/D$
n	$D/Q = 1/T$	
T^*	$\sqrt{\frac{2S}{DH}}$	$\sqrt{\frac{2S}{DH(1-D/P)}}$
n^*	$\sqrt{\frac{DH}{2S}}$	$\sqrt{\frac{DH(1-D/P)}{2S}}$
ρ util.	0	$D/P = T_p/T$ (need $D < P$)

- At Q^*: ordering cost = holding cost = TVC$^*/2$. EPQ downtime/yr $= nT_d$. Savings $= 1 - \text{TVC}(Q^*)/\text{TVC}(Q_{\text{current}})$.
- **Reorder point** (on-hand trigger, D, L same unit). $L \le T$: ROP $= DL$. $L > T$: inventory position still triggers at DL, but $\lfloor L/T \rfloor$ orders are in transit:
$$\text{ROP}_{\text{on-hand}} = DL - \lfloor L/T \rfloor Q$$
- EPQ: DL is the on-hand trigger only if $L \le T_d$; if $T_d < L \le T$ the reorder falls during the run and on-hand $= (P-D)(T-L)$.

2 · Sensitivity and robustness

$$r = \frac{\text{TVC}(Q)}{\text{TVC}(Q^*)} = \frac{1}{2}\left(\frac{Q^*}{Q} + \frac{Q}{Q^*}\right) \qquad \%\text{ increase} = (r-1)\times 100$$

- Steps: compute $Q^* \to$ take the other Q (given / rounded / policy) $\to r \to (r-1)100\%$.
- Benchmarks: $Q/Q^* = 4/3$ or $3/4 \to +4.2\%$ · $\sqrt{2} \to +6.1\%$ · 2 or ½ → +25%. TVC is flat near Q^*.
- **Scaling:** $Q^* \propto \sqrt{DS/H}$, TVC$^* \propto \sqrt{DSH}$, $n^* \propto \sqrt{DH/S}$. A parameter $\times k$ → multiply by $\sqrt{k}$ if it is in the numerator, divide by $\sqrt{k}$ if in the denominator. E.g. $S \times 4$: Q^* and TVC* ×2. To halve the lot, S must fall to $S/4$.
- **Mis-estimated input** (D or S off by factor k, or H off by $1/k$): $Q = \sqrt{k}Q^*$, penalty $r = \frac{1}{2}\left(\sqrt{k} + 1/\sqrt{k}\right)$ (e.g. $k = 2 \to +6.1\%$).

3 · Power-of-two policies

Variables: T^* optimal interval · T_B base period (1 week = 1/52 yr) · $T = 2^k T_B$, $k = 0, 1, 2, \ldots$ · $\text{TVC}(T) = S/T + HDT/2$ (with $Q = DT$).

1. $T^* = Q^*/D$ (convert to base periods).
2. **Range test:** choose the k with $2^k T_B \in \left[T^*/\sqrt{2},\ \sqrt{2}T^*\right]$ (exactly one exists).
   Equivalent: smallest k with $\text{TVC}\left(2^k T_B\right) \le \text{TVC}\left(2^{k+1}T_B\right)$.
3. Cost ratio $r = \frac{1}{2}\left(\frac{T^*}{T} + \frac{T}{T^*}\right) \le \frac{1}{2}\left(\sqrt{2} + \frac{1}{\sqrt{2}}\right) = 1.0607 \to$ at most +6.1%.
4. Lot $Q = DT$; % increase $= (r-1) \times 100$.

Table trick: list k, 2^k, "in range?" and tick the single yes. With several products, each gets its own 2^{k_i} on a common base → they synchronize.

4 · All-unit quantity discounts

Variables: Break points $q_0 = 0 < q_1 < \ldots < q_m = \infty$ · if $q_i \le Q < q_{i+1}$ **every** unit costs C_i · $C_0 \ge C_1 \ge \ldots$ · $H_i = hC_i$ changes with price.

$$\text{TC}_i(Q) = \frac{D}{Q}S + \frac{Q}{2}hC_i + DC_i \qquad q_i \le Q < q_{i+1} \qquad Q_i = \sqrt{\frac{2DS}{hC_i}}$$

1. Start at the **lowest** price. If Q_i is feasible ($q_i \le Q_i < q_{i+1}$): candidate Q_i, **stop** (higher prices cannot beat it).
2. If $Q_i < q_i$: candidate $= q_i$ (the break point); move to the next higher price and repeat.
3. Q^* = candidate with the lowest TC$_i$. Cycle inventory $= Q^*/2$.

Must include purchase cost DC_i (it differs across ranges): compare TC, not TVC. Read break points carefully: "1,001–2,000" → $q_i = 1\,001$.

5 · Aggregation (joint ordering)

Variables: m products · S common cost per order (truck/transport) · s_i product-specific cost (paid when i is in the order) · D_i, C_i, h, $H_i = hC_i$ · n orders/yr.

No aggregation (each product alone, $S_i' = S + s_i$)

$$Q_i^* = \sqrt{\frac{2D_iS_i'}{H_i}} \qquad n_i^* = \sqrt{\frac{D_iH_i}{2S_i'}} \qquad \text{TVC}_{\text{no}} = \sum_i \sqrt{2D_iS_i'H_i}$$

Complete aggregation (all products every order, $S' = S + \sum_i s_i$)

$$n = D_i/Q_i \text{ (same for all)} \qquad \text{TVC}(n) = nS' + \frac{\sum_i H_iD_i}{2n}$$

$$n^* = \sqrt{\frac{\sum_i D_iH_i}{2S'}} \qquad Q_i^* = \frac{D_i}{n^*} \qquad \text{TVC}_{\text{agg}} = \sqrt{2S'\sum_i D_iH_i} = 2n^*S'$$

- Savings $= \left(1 - \text{TVC}_{\text{agg}}/\text{TVC}_{\text{no}}\right) \times 100\%$.
- **Capacity check:** load per order $\sum_i Q_i^* = \sum_i D_i/n^* \le$ cap. If violated: $n = \sum_i D_i/$ cap ($> n^*$), $Q_i = D_i/n$, cost from TVC(n) above (no longer $2nS'$).
- Same check for no aggregation: if $Q_i^* >$ cap, set $Q_i =$ cap.
- Report per product: Q_i, n, cycle inventory $Q_i/2$, days of inventory $= Q_i/(2D_i) \times 365$.

6 · Newsvendor (single period)

Variables: D demand, $N(\mu, \sigma)$ · Q order qty · p price · c unit cost · s salvage per leftover · g goodwill per unit short (0 if not given) · C_u underage (1 too few) · C_o overage (1 too many) · α critical ratio · F CDF · Φ, φ std. normal CDF, pdf · $L(z)$ std. normal loss.

Order Q once, before demand → pay cQ → demand → sell $\min(Q, D)$ at p → leftovers at s → each shortage costs g. If unsure of C_u, C_o: profit lost by having 1 fewer / 1 more unit.

$$C_u = p - c + g \qquad C_o = c - s$$
$$\text{Marginal unit:}\quad C_oF(Q) = C_u(1 - F(Q))$$
$$F(Q^*) = P(D \le Q^*) = \alpha = \frac{C_u}{C_u + C_o} = \frac{p - c + g}{p - s + g}$$
$$z^* = \Phi^{-1}(\alpha), \quad Q^* = \mu + z^*\sigma$$
$$\text{round } Q \text{ up} \cdot \alpha < 0.5 \Rightarrow z^* < 0 \Rightarrow Q^* < \mu$$

Performance for any Q (first: $z = (Q - \mu)/\sigma$)

In-stock (Type I SL)	$P(D \le Q) = \Phi(z)$
Stockout prob.	$P(D > Q) = 1 - \Phi(z)$
E[short] (lost sales)	$E(D-Q)^+ = \sigma L(z), \quad L(z) = \varphi(z) - z(1 - \Phi(z))$
E[sales]	$\mu - E$[short]
E[left]	$Q - E$[sales] $= Q - \mu + E$[short] $= \sigma(z + L(z))$
Fill rate (Type II)	E[sales]$/\mu = 1 - E$[short]$/\mu$ (usually > in-stock)
Expected profit	$\pi = pE$[sales] $+ sE$[left] $- gE$[short] $- cQ$
	$= (p-c)\mu - C_uE$[short] $- C_oE$[left]
Max profit	$\pi(Q^*) = (p-c)\mu - (C_u + C_o)\varphi(z^*)\sigma$

- **Target in-stock** α: $Q = \mu + \Phi^{-1}(\alpha)\sigma$. **Target fill rate** FR: E[short] $= (1 - \text{FR})\mu \to L(z) = E$[short]$/\sigma \to z$ (larger one; L falls with z) $\to Q$.
- **Discrete demand table:** Q^* = smallest Q with $F(Q) \ge \alpha$ (cumulate probabilities). E[short] $= \sum_{d>Q}(d - Q)P(d)$; then the same chain.
- **Mismatch cost:** $C(Q) = C_uE$[short] $+ C_oE$[left], $C(Q^*) = (C_u + C_o)\varphi(z^*)\sigma$ → π^* is linear in σ; α does not depend on σ. Value of cutting σ (max to pay for a forecast tool) $= (C_u + C_o)\varphi(z^*)\Delta\sigma$.

Variants

- **Goodwill:** $C_u = p - c + g$ and $-gE[\text{short}]$ in π. Compare scenarios with the same g.
- **2nd (emergency) supplier** at c_P after demand ($c < c_P < p$): $C_u = c_P - c$, $C_o = c - s$. $\pi = p\mu + sE[\text{left}] - c_P E[\text{short}] - cQ$. Max fee for the option $= \pi_{\text{with}} - \pi_{\text{without}}$.
- **Fixed order cost** K: Q^* unchanged; profit $\pi(Q^*) - K$; order only if > 0 (max $K = \pi(Q^*)$).
- **Sensitivity:** $s \uparrow \Rightarrow C_o \downarrow \Rightarrow Q^* \uparrow \cdot\ g \uparrow$ or $p \uparrow \Rightarrow Q^* \uparrow \cdot\ c \uparrow \Rightarrow Q^* \downarrow \cdot\ \sigma \uparrow$: $Q^* \uparrow$ if $z^* > 0, \downarrow$ if $z^* < 0$; $\pi^* \downarrow$ always.

7 · Supply-chain contracts (double marginalization)

Variables: c supplier's production cost · w wholesale price · p, s, g retailer's price, salvage, goodwill · b buy-back price per return · r retailer's cost per return · v supplier's salvage per returned unit (0 if scrapped) · π_R, π_S, π_{SC}.

Each firm decides with its own margin → retailer uses $C_u = p - w < p - c$, $C_o = w - s > c - s$ → orders **less** than the SC optimum. Contracts shift risk/margin to coordinate. Every profit uses the same chain: $Q \to z \to L(z) \to E[\text{short}] \to E[\text{sales}] \to E[\text{left}]$. **Order:** solve the retailer first (its Q, $E[\text{left}]$), then plug them into π_S.

Contract	Critical ratio and profits
Wholesale	$C_u = p - w + g$, $C_o = w - s \to Q_R$ $\pi_R = pE[\text{sales}] + sE[\text{left}] - gE[\text{short}] - wQ$; $\pi_S = (w - c)Q$
Integrated SC	$C_u = p - c + g$, $C_o = c - s$ (w cancels) → larger Q $\pi_{\text{SC}} = pE[\text{sales}] + sE[\text{left}] - gE[\text{short}] - cQ$
Buy-back	$C_u = p - w$, $C_o = w - b + r$, $\alpha_R = (p - w)/(p - b + r)$ $\pi_R = pE[\text{sales}] + (b - r)E[\text{left}] - wQ$ $\pi_S = (w - c)Q - (b - v)E[\text{left}]$

- Loss from double marginalization $= \pi_{\text{SC}}^{\text{int}} - (\pi_R + \pi_S)_{\text{wholesale}}$.
- **Coordinate:** set the retailer's ratio equal to the SC ratio:

$$\frac{p - w}{p - b + r} = \frac{p - c}{p - s}$$

$$\text{given } w: \quad b^* = p + r - \frac{(p - w)(p - s)}{p - c}$$

$$\text{given } b: \quad w^* = p - \frac{(p - c)(p - b + r)}{p - s}$$

- Buy-back $\% = b/w$. Feasible: $b < w$ (no profit from ordering to return) and $b - r > s$ (returning beats salvaging), i.e. $s + r < b < w$.
- $\pi_R + \pi_S = pE[\text{sales}] + (v - r)E[\text{left}] - cQ$: equals the integrated π_{SC} only if $v - r = s$; otherwise slightly below even when coordinated.
- Accepted only if **both** π_R and π_S beat their wholesale-contract values.

Buy-back drawbacks: needs reverse logistics, admin cost, may push retailer to overstock low-value items. Other coordinating contracts: **revenue sharing** (low w + share of revenue), **quantity flexibility** (full refund up to a return quota), **quantity discounts** (induce larger orders). With goodwill, add $+g$ to every C_u and to both denominators.

8 · Order-up-to (periodic review, backorders)

Variables: L lead time (periods) · T review period ($T = 1$ default) · μ, σ demand per period · D_{L+1} demand over $L + 1$ periods · S order-up-to (base-stock) level · α target in-stock · h holding/unit/period · b backorder penalty/unit · N periods · K cost per order.

Long-life product, stationary random demand, unlimited replenishments, backorders. Each period: order → the order from L periods ago arrives → demand. **Inventory level** = on-hand − backorder; **inventory position** = level + on-order = S right after ordering. Order = S − position = last period's demand (one-for-one, pull). Level at end of period = $S - D_{L+1}$.

$$\mu_{L+1} = (L + 1)\mu \qquad \sigma_{L+1} = \sqrt{L + 1}\,\sigma \quad \text{(periods independent)}$$

$$S = \mu_{L+1} + z\sigma_{L+1}, \quad z = \Phi^{-1}(\alpha) \quad \text{or cost-based:} \quad \alpha = \frac{C_u}{C_u + C_o} = \frac{b}{b + h}$$

Given S	$z = (S - \mu_{L+1})/\sigma_{L+1}$ (recompute with the **rounded** S)
In-stock	$P(D_{L+1} \le S) = \Phi(z)$; stockout $= 1 - \Phi(z)$
$E[\text{backorder}]$	$\sigma_{L+1}L(z)$ (≈ newsvendor lost sales)
$E[\text{on-hand}]$ = SS	$S - \mu_{L+1} + E[\text{BO}] = \sigma_{L+1}(\varphi(z) + z\Phi(z))$
$E[\text{on-order}]$	$L\mu$ (Little's law; uses L, not $L + 1$; indep. of σ)
Fill rate	$1 - E[\text{BO}]/\mu$ (one period's demand, not μ_{L+1})
Cycle inventory	$\mu/2$
Days of supply	$E[\text{on-hand}]/\mu = \sqrt{L + 1}\,\dfrac{\sigma}{\mu}(\varphi(z) + z\Phi(z))$
Stockouts in N	$N(1 - \alpha)$; periods until the 1st $= 1/(1 - \alpha)$

- **Target fill rate** FR: $E[\text{BO}] = (1 - \text{FR})\mu \to L(z) = E[\text{BO}]/\sigma_{L+1} \to z$ (larger) → S.
- h in $b/(b + h)$ must be **per period** (annual $h \to h/52$ for weeks). Optimal in-stock is usually high ($b \gg h$).
- Safety-stock drivers: lead time, variability σ/μ, service level. $L \downarrow \Rightarrow \mu_{L+1}, \sigma_{L+1}, S$, on-order and SS $\downarrow$ (strongest at high α).

General review period T (replace $L + 1$ by $L + T$)

$$\mu_{L+T} = (L + T)\mu \quad \sigma_{L+T} = \sqrt{L + T}\,\sigma \quad S = \mu_{L+T} + z\sigma_{L+T}$$

$$E[\text{BO}] = \sigma_{L+T}L(z) \quad \text{SS} = S - \mu_{L+T} + E[\text{BO}] \approx z\sigma_{L+T}$$

- Cycle inventory $= \mu T/2$ · average inventory = SS $+ \mu T/2$ · pipeline $= L\mu$ · fill rate $= 1 - E[\text{BO}]/(\mu T)$.
- Days of cover: SS$/\mu$ and cycle $T/2$. Orders/yr $n = (1 - P(D_T = 0))/T_{\text{yr}}$ ($= 1/T_{\text{yr}}$ for normal demand).
- **Annual cost** = avg. inventory × H ($+L\mu H$ if the firm owns goods in transit) $+nK$ (+ transport/unit × D).
- **Choosing T:** lowest ordering + holding cost. $T \uparrow \to$ fewer orders, more SS and cycle stock (EOQ-like trade-off). Keep all costs in the same time unit.

9 · Normal-table rules (tables provided separately)

- $L(z) = \varphi(z) - z(1 - \Phi(z)) \cdot\ \Phi(-z) = 1 - \Phi(z) \cdot\ \varphi(-z) = \varphi(z) \cdot\ L(-z) = L(z) + z \cdot\ z + L(z) = \varphi(z) + z\Phi(z)$.

Rounding: $\alpha \to z$: round-up rule (take the larger table z; use all decimals, e.g. $\Phi(2.43) = 0.99245 < 0.9925 \Rightarrow 2.44$) · $x \to z = (x - \mu)/\sigma$ to 2 decimals · Q, ROP, S, newsvendor Q rounded **up** (EOQ: normal rounding) · expected values not rounded.
Wording: "at least / above" → $1 - \Phi(z)$; "at most / below" → $\Phi(z)$ · a probability → Φ; an expected quantity → $L(z)$.

10 · Traps and check values

- **Units:** D, H per the same year; P, D same rate; L, T, μ, σ same period; σ scales with $\sqrt{\text{time}}$, never linearly.
- **EOQ/EPQ:** with discounts $H_i = hC_i$ changes per price · EPQ average inventory is $I_{\max}/2 \cdot L > T$: subtract pipeline orders from ROP.
- **Discounts:** compare TC **with** DC_i; a break point is a candidate only when the EOQ falls below it.
- **Aggregation:** S' includes **every** s_i · check truck capacity.
- **Newsvendor:** add g to C_u (not to C_o) · $\alpha < 0.5$ gives negative z · fill rate > in-stock prob. · $E[\text{left}]$ uses Q, $E[\text{sales}]$ uses μ.
- **Contracts:** buy-back $C_o = w - (b - r)$, not $w - s$ · solve the retailer first, supplier second · % buy-back $= b/w$.
- **OUT:** S covers $L + 1$ (or $L + T$), on-order uses L · fill rate divides by one period's demand · h per period in $b/(b + h)$ · recompute z with the rounded S.

Check values. Newsvendor (Hammer 3/2): $\mu = 3192$, $\sigma = 1181$, $p = 190$, $c = 110$, $s = 90 \to C_u = 80$, $C_o = 20$, $\alpha = 0.8$, $z = 0.85$, $Q^* = 4196$.
Contracts (ski jackets): $\mu = 1000$, $\sigma = 300$, $p = 250$, $c = 100$, $s = 75$, $w = 175 \to$ retailer $\alpha = 0.429$, $z = -0.18$, $Q = 946$, $\pi_R \approx 54\,392$, $\pi_S = 70\,950$ ($\pi_{\text{SC}} = 125\,342$). Integrated: $\alpha = 0.857$, $z = 1.07$, $Q \approx 1321$, $\pi_{\text{SC}} \approx 138\,154$ (+10%). Buy-back with $r = 5$: $b^* = 167.5$.